Grading Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Fruit grader; Grading Machine

What it is used for

A grading machine sorts fruits or other suitable food products by size, weight, colour, or another specified quality parameter. Use the correct grading settings and handling method to reduce product damage and maintain consistent output.

Main parts

- Feed hopper or loading conveyor
- Sorting belt, rollers, screens, or grading channels
- Size, weight, colour, or optical sensors, if fitted
- Drive motor and transmission
- Control panel and adjustment controls
- Collection bins or discharge conveyor

Before you start

1. Confirm the machine is suitable for the product being graded.
2. Inspect belts, rollers, screens, guards, sensors, and collection bins.
3. Clean and sanitise food-contact surfaces according to the hygiene procedure.
4. Check the emergency stop, guards, and other safety devices.
5. Set grading criteria and verify the relevant sensors or measuring system.
6. Remove foreign objects and ensure collection bins are clean and correctly positioned.

How to use

1. Keep the machine switched off while cleaning or adjusting internal parts.
2. Install the required screens, guides, or approved product-handling attachments.
3. Set the grading parameters according to the product specification.
4. Position clean collection bins at the correct discharge points.
5. Close guards and confirm that the feed and discharge paths are clear.
6. Power on the machine and run it empty briefly to check normal operation.
7. Feed the product gradually at the recommended rate.
8. Observe the sorting process and check that products move freely without excessive impact.
9. Inspect samples from each grade for consistency and visible damage.
10. Adjust settings only according to the approved procedure if grading is inaccurate.
11. Stop feeding and allow the machine to clear the product flow before shutdown.
12. Switch off and isolate the power before removing jams or cleaning internal parts.
13. Collect and label graded products according to the workplace procedure.

4. Clean the machine and record faults or unusual grading results.

⚠ Safety

- Only trained operators should operate or adjust the machine.
- Keep hands and tools away from moving belts, rollers, screens, and drive components.
- Never bypass guards, interlocks, or emergency stops.
- Isolate and verify zero energy before clearing jams or performing maintenance.
- Use safe lifting methods when handling heavy product bins.
- Follow food hygiene requirements and keep cleaning chemicals away from exposed products.

Care & storage

- Clean and sanitise food-contact parts after the production run.
- Remove product debris from belts, rollers, screens, and sensors.
- Inspect belts, bearings, fasteners, guards, and drive components regularly.
- Calibrate or verify grading sensors according to the maintenance schedule.
- Lubricate only specified points and keep lubricant away from food-contact surfaces.

Common mistakes to avoid

- Feeding products too quickly and causing jams or inaccurate grading.
- Using incorrect grading settings for the product batch.
- Ignoring dirty sensors, worn belts, or damaged screens.
- Clearing jams while the machine remains energised.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.